Department of Economics

MSc Extended Essay Cover Sheet

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TITLE	**Irrational markets: how disagreements and backwards looking expectations impact the New Keynesian model (Structured Essay)**

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Irrational markets: how disagreements and backwards looking expectations impact the New Keynesian model

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1 Introduction

The modern central bank cannot simply rely on conventional monetary policy tools to tame inflation. It is well acknowledged that future inflation expectations play a major role in determining the inflation rate of the present. However, merely setting the overnight rate has little impact on the long run. To influence the future path of interest rates, central banks need to talk. In particular, financial markets play a key role in enabling the successful transmission of monetary policy communication. In 2012, in the middle of the Euro debt crisis, ECB president Mario Draghi's infamous words "Whatever it takes [...] and believe me it will be enough" caused the Italian bonds spread to move by 22 bps in the days following the speech (Krishnamurthy et al., 2018). Prices of long maturity assets reflect participants' views of future economic conditions, including the central bank's policy rate. Through speeches, memos and press conferences, policymakers can anchor market expectations and influence the long end of the yield curve. As such, understanding the channels through which central bank communication can affect financial markets expectations is crucial to modern monetary policy.

However the core of current macroeconomic models does not adhere to the complexities of the task faced by central banks. The most common modelling approach relies on the New Keynesian model (Walsh, 2017). Whilst the model allows for changes in nominal monetary variables to affect real variables (Galí, 2018), communication has no role to play in this setting. This is partly due to the fact that all agents are assumed to have rational expectations (RE). However, as stated by former Bank of England governor Mervyn King, the rational behavior assumption is not a good fit for expectations formation in the real world (King, 2012). Evidence shows that financial markets can disagree with central banks (Couture, 2021). On the other

hand, adding complexity in the form of more realistic assumptions about expectations can lead to intractable models. These often require difficult mathematical techniques to solve and do not always provide more insights into the phenomena we are trying to understand.

This paper aims to evaluate whether adding more complex assumptions about markets expectation formations to the classical New Keynesian model provides any additional insights for central bank communication. To achieve this we compare two alternative assumptions based on the work of Caballero and Simsek, 2022. Firstly, the market and the central bank can disagree on the state of aggregate demand and other variables such as inflation and the output gap. Secondly, agents' expectations will be a weighted average of their expectations in the previous period, the state of aggregate demand in the previous period and the economic signals of the current period. We seek to enrich Caballero and Simsek, 2022's paper by providing a comparison between different combinations of these two assumptions. In one version we model both assumptions, replicating the work of Caballero and Simsek, 2022 with some minor simplifications. A second version evaluates outcomes with just disagreements. The final version has only common beliefs with the backwards-looking expectations assumption. We will compare each model with the empirical findings of Christiano et al., 2005 for the United States as well as the representative agent New Keynesian model as described in Walsh, 2017.

The sections of this essay are arranged as follows. Section 2 contains a literature review of monetary policy models without rational expectations, Bayesian expectations formation and disagreements between central banks and financial markets. Section 3 provides a description of our three models: the simplified version of the Caballero and Simsek, 2022 model, a second version without the adaptive learning expectations and a third without disagreements. Section 4 provides an analysis of the

results. We plot impulse response functions (IRFs) for inflation, the output gap, the nominal interest rate as well as both agent's expectations. We find that the original model replicates some of the patterns in the data. Disagreements introduce some persistence and variability in the responses of output and inflation. However most of the differences from the New Keynesian model arise because of the Bayesian expectations assumption. We also provide some normative conclusions for central bank communication. In Section 5 we discuss the limitations in our approach as well as a potential next step.

2 Literature Review

This analysis contributes to the literature about the New Keynesian model (for an overview see Galí, 2018). It builds upon a classical version of the model with monetary policy discretion as in Walsh, 2017. The essay also contributes to the study of modeling monetary policy without rational expectations (see Woodford, 2013 for a review). A fraction of the literature focuses on the inattention of private agents as a departure from rational expectations. In Gabaix, 2020, private agents are partially myopic and do not anticipate future events whilst Angeletos and Lian, 2018 and Angeletos and Huo, 2021 explore the macroeconomic effects of dispersed information at the micro level. In our comparison we assume both market participant and policy makers consider all the information available. Non-rational expectations arise due to different interpretations of the same information set. Other papers explore different variations of bounded rationality. For instance Farhi and Werning, 2019 assume a "level-k thinking" where agents perceive policy announcements correctly but cannot foresee its implications. Angeletos and Sastry, 2021 has the same approach but private agents fail to understand how the partial equilibrium affects the general equilibrium.

In our model, markets fully understand the mechanisms of the equilibrium as well as the central bank decision rule. Disagreements arise due to different exogenous opinions about aggregate demand.

Our evaluation of the opinionated markets model also contributes to a vast literature about adaptive learning and its implications for monetary policy. Bernanke, 2004 states that the adaptive nature of inflation expectations can be an added source of macroeconomics volatility. In Evans and Honkapohja, 2003 private agents have backwards-looking learning rule and inflation is unstable if monetary policy does not take these views into account when setting the nominal interest rate. In our model, we adopt a similar framework. Both the market and the monetary authority form their expectations based on a weighted average of past experiences and new information. The market's expectations are taken into account in the nominal interest rate set by the central bank. In Bordalo et al., 2018 agents beliefs also evolve with observed data. However, the biases in their judgement is heavily focused on the hypothetical future whereas we only allow expectations to be influenced by current and past information of the economy. Recent empirical work confirms Bayesian behavior is observed in the data. Cavallo et al., 2017 show that even when new information is available, individuals still place a significant weight on inaccurate sources of information such as their memories of the previous periods. McMahon and Rholes, 2023 show through survey data that individuals exhibit recency bias when evaluating a central bank economic forecast. Bordalo et al., 2020 finds evidence of an overreaction to economic news in the data. We model markets who don't necessarily overreact but will adapt their expectations to news or signals of the economy.

This essay also relates to previous studies on central bank and financial markets disagreements and their implications for monetary policy (see Simsek, 2021 for a review). Empirical studies show that observed disagreements between financial markets

and central banks have predictive power for future monetary policy(Giacoletti et al., 2021). Our disagreements assumption models this phenomenon. Disagreements between markets and policy makers drive expectations which in turn affect aggregate demand, aggregate supply and monetary policy. Andrade and Ferroni, 2021 shows that news about macroeconomic variables can affect the market reaction to monetary policy. In our assumptions, signals about aggregate demand and their respective interpretation are key drivers of inflation and monetary policy. The theoretical model in Stein and Sunderam, 2018 has a central bank with private information as well as aversion to bond market volatility. In our model, both agents agree to disagree and volatility can only arise through opposing views. Caballero and Simsek, 2020 focuses on the links between an overshooting of monetary policy and the discrepancy between asset prices and the real economy. We will consider policy overshoots and undershoots but we will not extend the analysis to asset pricing.

Finally, our analysis aims to provide insights about the effects central bank communication on financial markets. It this context we can relate it to the large volume of studies on the effects of central bank communication on financial markets' expectations and outcomes (see Blinder et al., 2008 for a comprehensive summary). A large volume of empirical papers show that long-run interest rates and output gap expectations react to monetary policy speeches and announcements (Romer and Romer, 2000, Andersson et al., 2006 andNakamura and Steinsson, 2013). Our analysis of different market expectations assumptions seeks to help explain the behavior behind these findings. On the theoretical side, Melosi, 2017 presets a DSGE model where propagation of shocks is highly dependent on how price-setters interpret monetary policy signals. In our comparison, markets have their own interpretation of economic signals. However its interpretation still impacts monetary policy. In Macaulay, 2022 heterogeneous expectations affect the transmission of macroeconomic shocks. We

don't find different beliefs to be particularly significant in the reaction to a demand shock but it contributes to persistence in the system.

3 The Model

3.1 The Baseline Model

We propose a simplified version of the model developed by Caballero and Simsek, 2022. We start from the baseline of a standard representative agent New Keynesian model (Walsh, 2017). The market adopts the role of the household and makes consumption and labor decisions. The supply side of the economy is divided into a competitive final goods sector and a intermediate goods sector. the latter displays the classic monopolistic competition assumption. The model follows the Calvo pricing framework. In any given period, a randomly selected fraction θ of intermediate goods firms do not update their prices. In equilibrium, the goods market clears and we observe $Y_t = C_t$. This set up allows us to derive the classical New Keynesian Phillips Curve:

$$\pi_t = \beta \mathbb{E}_t^m(\pi_{t+1}) + \kappa \widetilde{y}_t \tag{1}$$

The subscript m on the expectations denotes the market's expected inflation which is allowed to differ from the central bank's inflation expectations. This is the first departure from rational expectations. In traditional models, we assume that all agents have access to the same information set. They form identical beliefs about future macroeconomic variables. Here we assume they will have opposing views on the economy. This is due to exogenous interpretations of the same facts and not because of asymmetric information. Both parties have full visibility into the other's optimisation procedure, the general equilibrium and the set of prices in the economy. So they

simply agree to disagree.

We allow for aggregate demand shocks through fluctuations in the growth rate of productivity between period t and $t+1$. Productivity a_t follows the process:

$$a_{t+1} = a_t + g_t \tag{2}$$

The growth rate g_t is realized at the end of period t. So at the beginning of the period g_t is a stochastic variable on which agents form beliefs. Similarly, agents will not know the realisations of inflation and the output gap until the end of period t.

From Euler equation we can derive the IS curve:

$$\widetilde{y}_t = -(\widetilde{i}_t - \mathbb{E}_t^m[\pi_{t+1}] - \rho) + g_t + \mathbb{E}_t^m[\widetilde{y}_{t+1}]) \tag{3}$$

where $\rho = \log(\beta)$ and $r_t = i_t - \mathbb{E}_t^m[\pi_{t+1}]$ is the real interest rate under market expectations. The equations tells us that a one unit increase in the productivity growth rate g_t will increase the output gap by one unit. As such, Caballero and Simsek, 2022 also refer to g_t as the aggregate demand shock.

Monetary policy sets the interest rate i_t. To model transmission lags, the rate is set at the beginning of the period, before observing aggregate the aggregate demand shock for period t as well as π_t and $\widetilde{y}_t$. We assume the central bank sets the interest rate with discretion and solves the following optimization problem:

$$\min_{i_t} \mathbb{E}_t^{cb}[\gamma\widetilde{y}_t^2 + \pi_t^2] + \mathbb{E}_t^{cb}[\mathbb{V}_{t+1}^{cb}] \text{ subject to (1) and (3)} \tag{4}$$

where $\mathbb{V}_{t+1}^{cb} = \sum_{h=1}^{\infty} \beta^h(\gamma\widetilde{y}_{t+h}^2 + \pi_{t+h}^2)$. Again we emphasize the cb subscript on the expectations of future inflation and output gap. This denotes the central bank's inflation expectations. As explained before it will be allowed to differ from those of

the market.

The aggregate demand is decomposed into a structural component demand G_t and a short term component of demand v_t:

$$g_t = G_t + v_t \qquad v_t \sim \mathcal{N}(0, \sigma_v^2) \tag{5}$$

$$G_t = G_{t-1} + \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, \sigma_\epsilon^2) \tag{6}$$

The term v_t represents transitory shocks to demand. The permanent component of demand G_t evolves following a random walk where ϵ_t captures exogenous changes to the growth rate of productivity. Both the central bank and the market do not observe g_t until the end of the period. They also cannot distinguish between G_t and v_t, they can only observe g_t. Let us assume agents care about the permanent component of demand G_t to make investment, savings and consumption decisions. Since both agents cannot observe G_t, they form expectations of the current realisation of G_t based on signals of the real state of the economy emitted at the beginning of each period. Caballero and Simsek, 2022 assume that the signals are equally observed by both agents. There is no inattention or private information. However, each agent has a different exogenous interpretation μ_t^j, $j = cb, m$ of the signal:

$$\mu_t^{cb}, \mu_t^m \sim \mathcal{N}(0, \sigma_\mu^2), \qquad \text{corr}(\mu_{cb}, \mu_m) = \rho_\mu \tag{7}$$

When $\rho_\mu = 1$, both agents have the same interpretation and no disagreements arise. When $\rho_\mu < 1$, they can disagree. This is the key building block that allows for different expectations to exist in the model. Agents have different exogenous interpretations of the underlying true state of the economy G_t. As this component of demand remains unobserved, disagreements can persist beyond period t, until there

is sufficient evidence to confirm one view or the other. The concept of persistence is captured by the parameter ψ, which we introduce below. We leave the derivations of the beliefs of both agents to Caballero and Simsek, 2022. By applying a Kalman filter, the authors show that agents update their beliefs of the structural component of demand according to a partially backwards looking learning process. Let $G_t^j = \mathbb{E}_t^j(G_t)$ for $j = m, cb$ denote agent's j, expectation of the structural component of demand. Then we state without proof:

$$G_t^j = \psi G_{t-1}^j + \omega_s(s_t + \mu_j) + \omega_g g_{t-1}, \tag{8}$$

for $\psi, \omega_s, \omega_g > 0$ and $\psi + \omega_s + \omega_g = 1$. Agent j's period t expectation of the permanent component of demand is a weighted sum of their previous period expectation of G_{t-1}^j, the observed signal of the state of the economy s_t, their unique interpretation of the signal μ_t^j and the realised aggregate demand shock in $t-1$. This is the model's second departure from rational expectations. In the canonical version of the New Keynesian model, agents fully understand and observe the equilibrium mechanisms and form beliefs about future variables based solely on information observed at time t. In short, there is no backward looking behavior. Here we allow for agents' beliefs of yesterday's structural part of demand to have some weight in their beliefs today. Similarly, letting agents draw their own interpretation of the signal s_t further amplifies the distinction with the RE case. Under rational expectations, both agents would only infer $\mathbb{E}_t^j(G_t^)$ based on s_t. In this instance they can deviate from the news according to their own exogenous beliefs.

Incorporating these belief dynamics into the previous equations we obtain the following closed form solutions for both agent's expectations of future inflation and

output gap. Further derivations can be found in Caballero and Simsek, 2022:

$$\mathbb{E}_t^m(\pi_{t+1}) = \Pi^m \psi(G_t^m - G_t^{cb}) \tag{9}$$

$$\mathbb{E}_t^{cb}(\pi_{t+1}) = \Pi^{cb} \psi(G_t^m - G_t^{cb}) \tag{10}$$

$$\mathbb{E}_t^m(\widetilde{y}_{t+1}) = \Gamma^m \psi(G_t^m - G_t^{cb}) \tag{11}$$

$$\mathbb{E}_t^{cb}(\widetilde{y}_{t+1}) = \Gamma^{cb} \psi(G_t^m - G_t^{cb}) \tag{12}$$

where $\Pi^{cb} = \frac{\gamma\beta\psi}{\gamma+\kappa^2-\gamma\beta\psi}\kappa$, $\Pi^m = \kappa + \Pi^{cb}$, $\Gamma^{cb} = -\frac{\kappa^2\beta\psi}{\gamma+\kappa^2-\gamma\beta\psi}$ and $\Gamma^m = \frac{(\gamma+\kappa^2)(1-\beta\psi)}{\gamma+\kappa^2-\gamma\beta\psi}$ are functions of the New Keynesian model parameters. The expressions above summarise the new insight of Caballero and Simsek, 2022: disagreements matter and give rise to different paths for the central bank's and the market's expectations of inflation and the output gap. These in turn drive inflation and the output gap through the Phillips curve and the IS curve (equations (1) and (3)).

Combining the closed form solutions together with the Fisher equation, we obtain the following expression for the real rate:

$$i_t = \rho + (1 - \widetilde{\psi})G_t^{cb} + \widetilde{\psi}G_t^m + \Pi^m \psi(G_t^m - G_t^{cb}) \tag{13}$$

Notice that we do not allow for monetary policy shocks in the expression for i_t. The market and the central bank will always disagree. Thus the optimal monetary policy will be shocking to the market and has the same effects as monetary policy shock. We note that the central bank will take into account the market's expectations when setting monetary policy. It can overshoot or undershoot monetary policy if the market expectation warrants it.

In our analysis we will compute the model above using Caballero and Simsek, 2022's calibrations. We leave the details to the appendix. Next we present the

different variations of the model.

3.2 Disagreements under Rational Expectations

Our first departure from Caballero and Simsek, 2022 is to convert the belief formations of both agents into a rational expectations framework. Under rational expectations we have:

$$G^m_{t+1} = G^m_t + \epsilon^m_t, \qquad \epsilon^m_t \sim \mathcal{N}(0, \sigma^2_m) \tag{14}$$

$$G^{cb}_{t+1} = G^{cb}_t + \epsilon^{cb}_t, \qquad \epsilon^{cb}_t \sim \mathcal{N}(0, \sigma^2_{cb}) \tag{15}$$

The two expressions above tell us that if no new information ϵ^m_t, ϵ^{cb}_t were to be observed at time t, agents' expectations of the demand shock would be unchanged between period t and period $t+1$. In this variant, agents still do not observe the permanent component of demand G_t. They form their respective expectations of exogenously. These expectations will vary across time following a random walk process. The differentiation between both agents will arise through different variances. Here we assume $\sigma^2_{cb} < \sigma^2_m$. This is to reflect the fact that markets update their views on the status of demand every day almost instantaneously and reveal them through active trading in several asset markets. Central banks will publicly update their views on demand more sparsely in a handful of press conferences and other communication events throughout the year. This specification will allow us to see if disagreements without the Bayesian updating assumption yield any insights or if most of the explanatory power of the model needs this additional assumption.

3.3 Common Beliefs

Next we revert back to the original assumption of Caballero and Simsek, 2022 of an updating learning structure for both agents' expectation formation process. However in this iteration, the central bank and the market will have the same beliefs about the structural state of the economy G_t for all t:

$$G_t^{com} = \mathbb{E}_t^m(G_t) = \mathbb{E}_t^{cb}(G_t) \tag{16}$$

$$G_t^{com} = \psi G_{t-1}^{com} + \omega_s(s_t + \mu_t) + \omega_g g_{t-1} \tag{17}$$

where the subscript *com* denotes the common beliefs. Here both agents draw the same interpretation μ_t from the public signal s_t. Equations (9)-(12) will be 0 under this assumption. Comparing this with the model described in the previous section will allow us to evaluate the contribution of each assumption.

4 Results

4.1 The Baseline Model

Let us now run our version of the Caballero and Simsek, 2022 model in dynare (Adjemian et al., 2024) with a high persistence calibration to obtain impulse response functions (IRFs) which allow us to evaluate the results of the model. We are interested in shocks to the structural component of demand G_t as these are the demand shocks defined in the original paper. We reiterate that this model does not include exogenous monetary policy shocks. The central bank's reaction to a demand shock will differ from the market's reaction due to idiosyncratic interpretations of the demand signals s_t. The optimal policy response according to the central bank's perspective will

be seen as a mistake by the market due to their opposing views. We can call it a 'monetary policy' shock. Therefore every demand shock in this model is followed by a monetary policy shock due to the presence of disagreements.

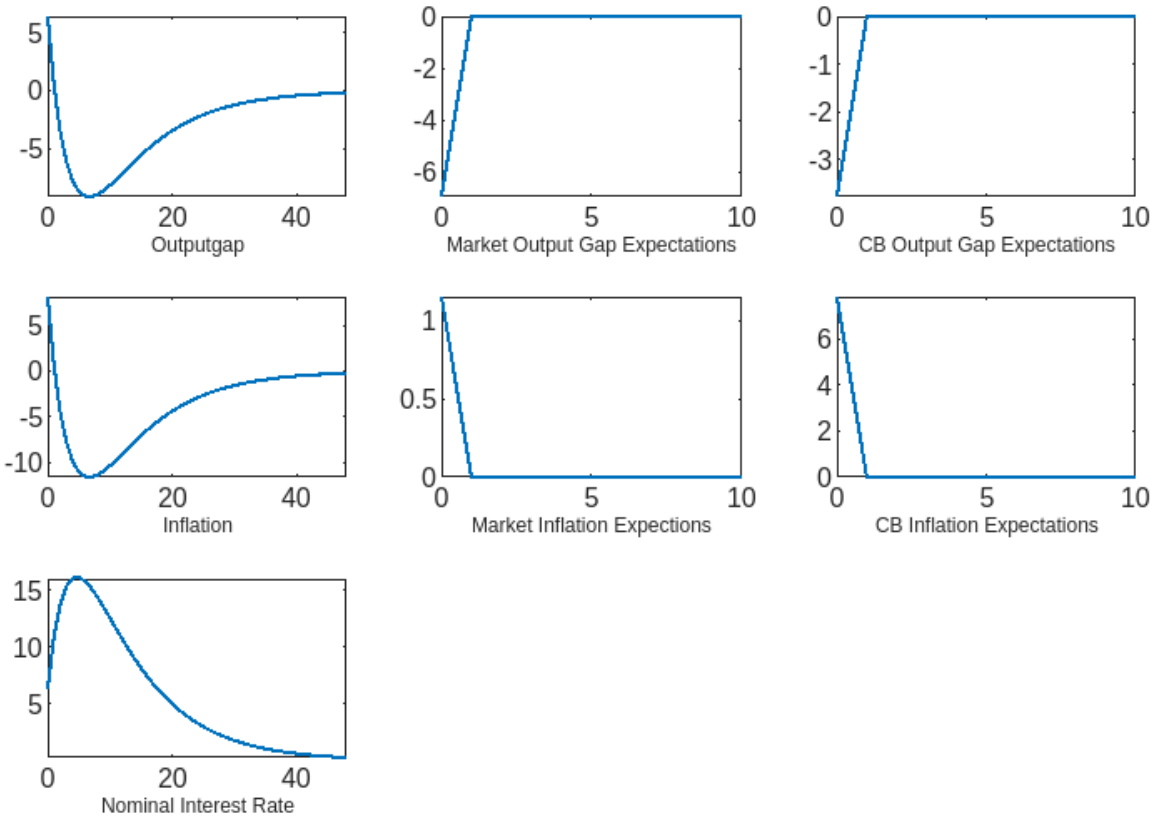

Figure 1: Impulse Response Functions for the Baseline Model with High Persistence Calibration

In our simplified replication of the Caballero and Simsek, 2022 model, we obtain inverted hump-shaped responses for inflation and the output gap to a positive demand shock. This is a desired property. Most empirical evidence for the United States shows that there is some inertia in inflation and other macro variables following a monetary policy shock (Christiano et al., 2005). Demand shocks are closely linked with monetary shocks in our model so we can conclude that the model replicates some of the empirical findings. The inertia is partly due to the backwards looking behavior

of firms, who update their views slowly. This feature is represented in the model via the Bayesian learning assumption in equation (8). However the IRF shapes obtained are different than what we see in the data. Following a demand shock, inflation and the output gap increase instantaneously in period t following a demand shock. Both variables then decrease immediately after, passing the steady state value and heading into deflation until they reach a minimum in period $t + 7$. The variables then slowly converge to steady state. In Christiano et al., 2005, the data shows a delayed response of output and inflation to the monetary policy shock. Output and inflation exhibit a positive hump shaped increase before declining, unlike our IRFs. The data does allow for variables to decrease past the steady state value but to less extreme values then the ones we see in the our results (in fact, just below 0). The data converges after 20 periods. Our model shows that convergence to the steady state takes longer, happening only after $t + 42$.

Looking at the agents expectations and at the mechanics of the model, we provide an interpretation for this pattern. Following the demand shock signal s_t, we can see that the central bank has a high inflation expectation. The market's inflation expectation is considerably lower. In this instance the market is less optimistic about the state of aggregate demand. Given its views, the central bank sets the nominal interest rate in period t under discretion with a trade off between output gap and inflation. So it is consistent to have a lower central bank expectation of the output gap compared to the market. We notice nominal interest rate i_t is set well and above the market's inflation expectation. The market is skeptic about the state of demand, thus $G_t^m < G_t^{cb}$. It follows from equation (13) that the central bank sets a lower interest rate than what it would otherwise do to accommodate for the market's views. However the market still deems this rate too high and it perceives the response as unfit given the signals. In other words, it perceives the response as a monetary

policy shock that will create a very negative output gap. The disagreement amplifies the monetary policy effect, which explains the quick decrease in inflation and output.

The market's and the policymaker's expectations of $\widetilde{y}_t$ and of π_t converge in less than one period to their steady state values. However there is inertia in the system due to the backwards looking behavior of their demand expectations. The beliefs in period t will be taken into account in their future demand expectations G^j_{t+1} as long as the persistence parameter ψ is strictly positive. This explains why the nominal interest rate keeps hiking even though inflation is on a downwards trajectory.

Running the model with a low persistence calibration, we see the same patterns for inflation and the output gap in Figure 2 below. Although in this instance, both variables fall more quickly after the initial shock and converge from period $t + 3$ onwards. We still lack the delayed initial response we find in the data. However the drop just below the steady state level and the subsequent convergence back up to the steady state do closely resemble the behavior of the plots in Christiano et al., 2005. However in the data convergence happens later at approximately 20 periods after the shock.

We note that in this case, the central bank expects positive inflation whilst the market expects negative inflation. The market is pessimistic about aggregate demand whilst the monetary authority is optimistic. The expectation of $\widetilde{y}_t$ from monetary policy is contradictory to the logic explained previously.

We conclude that initial disagreements are fundamental to create a monetary policy shock and amplify the actions of monetary policy compared to the classical New Keynesian model. Their propagation through time seems to be mostly due to the Bayesian expectations and more specifically due to the positive persistence of past beliefs.

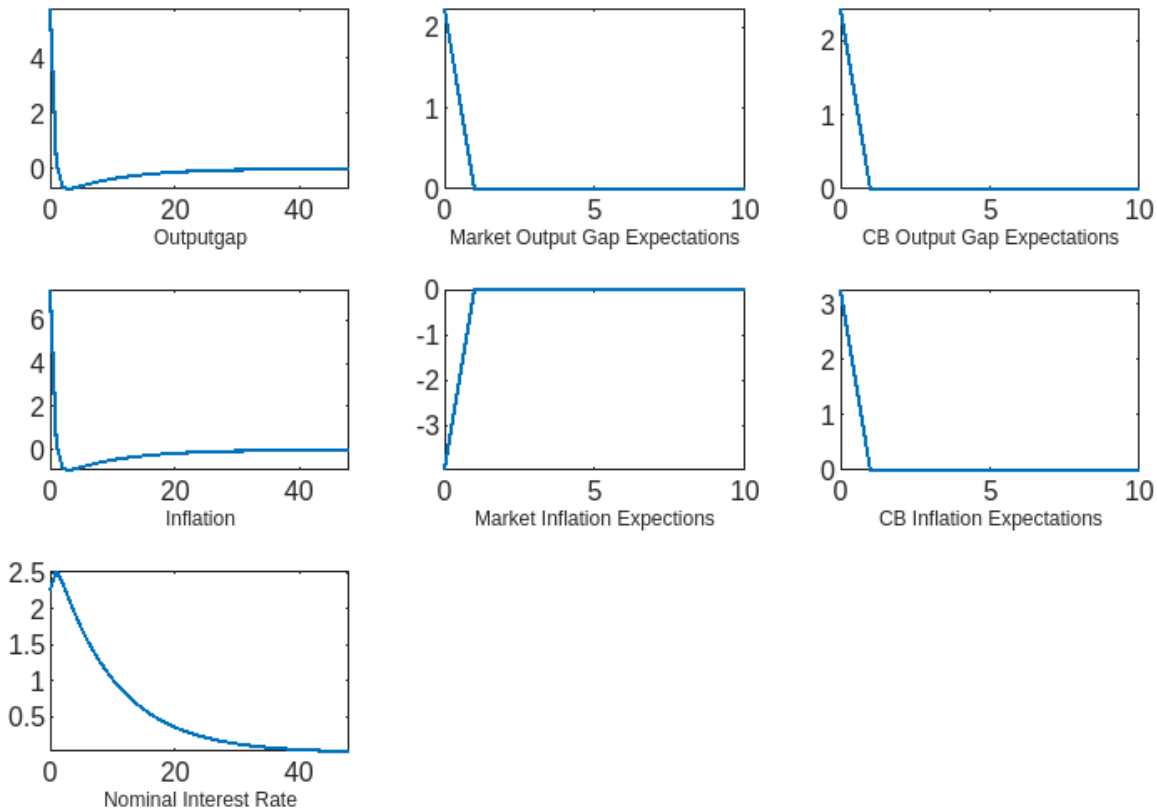

Figure 2: Impulse Response Functions for the Baseline Model with Low Persistence Calibration

4.2 Disagreements under Rational Expectations

We now take a closer look at the model with both rational expectations and disagreements. Figure 3 shows that we no longer have the inverse hump-shaped responses of $\widetilde{y}_t$ and of π_t we saw in the previous section. After a positive demand shock in period t, both variables increase instantaneously and immediately start to converge back to steady state. Although this sets us further away from the empirical findings of Christiano et al., 2005 the result is consistent with the IRFs obtained with the standard New Keynesian model (Walsh, 2017). The convergence happens at the same rate as in our baseline model. This is slower than what what we see in the New Keynesian model. Monetary policy is less powerful with the disagreements assumption in the

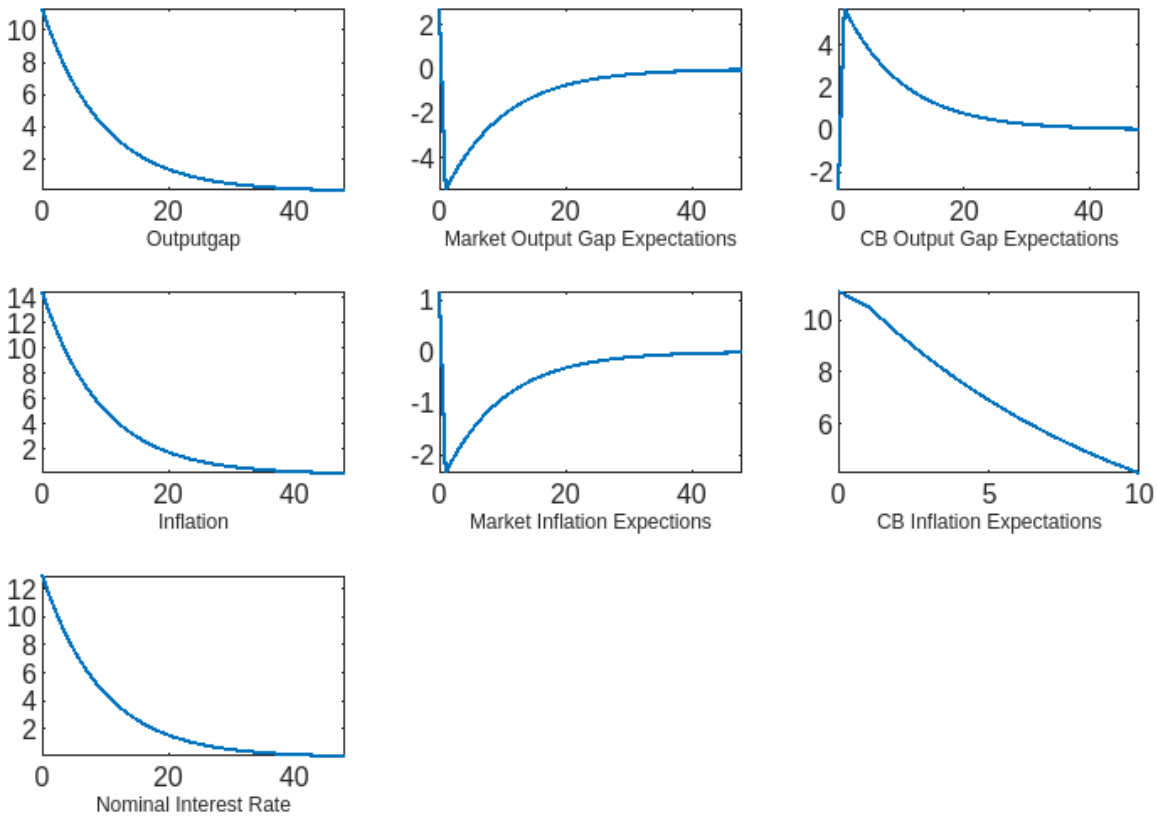

Figure 3: Impulse Response Functions for the RE Model with Disagreements

New Keynesian model.

The central bank's inflation expectations also exhibit the same response as π_t. However the market's inflation expectations increase in period t but drop suddenly after, plummeting into deflation. Similarly, the central bank's expectation of the output gap increases but with a lag and peaks in period $t+1$ before gradually returning to steady state. The market's expectation of the output gap expectation increases instantaneously at time t before declining into negative numbers at $t+1$. Afterwards they converge to the steady state. Although these last results seems counter intuitive, we can conclude that disagreements still arise in this set up. In this instance, the market is more optimistic about demand compared to the central bank $G_t^m > G_t^{cb}$. Disagreements exhibit some persistence and both agents' expectations take longer to

converge towards their steady state value.

The nominal interest rate i_t also increases but by more than what agents' expectations would suggest. We can explain this with equation (13). The market's optimism is cause for concern. Monetary policy takes into accounts the market's views and hikes the interest rate by more than what it would optimally decide in equilibrium. However witnessing i_t higher than their expectations induces the market to anticipate a recession. These expectations drive both the Phillips curve (1) and IS (3) curve and so we witness a sudden drop to a negative $\widetilde{y}_t$ and negative π_t.

Intuitively, we can suppose the the central bank is more attentive to the data and quickly assimilates its expectations to the signals it sees in the data. Markets are no as attentive to demand signals as before. In fact $s_t + \mu_t^m$ no longer appears in their expectations of demand in equation (14). As such, it takes longer to be convinced that monetary policy is being effective.

Although we do not witness the hump shaped reactions as before we still witness some persistence in macroeconomic variables. This persistence appears to be driven by the expectations disagreements between the market and the central bank. Some of the expectations formation behavior seems counter intuitive and warrants further exploration which we leave to future research.

4.3 Common Beliefs

Now we revert back to the adaptive learning assumption but both agents have the same expectations about demand G_t^j. We focus on just inflation and the output gap, as expectations of both agents are on the same path. In both the high and low persistence calibrations we observe the same IRFs shapes as in section (4.1).

Looking at the high calibration plots in Figure 4, we notice that both $\widetilde{y}_t$ and π_t

drop less drastically after their initial increase at time t. In fact, the output gap drops just below 0 in period $t+10$ before converging slowly back to steady state. In the opinionated markets version $\widetilde{y}_t$ decreases below -8 in period $t+7$. Inflation also drops to just -1 at $t+10$ compared to below -10 at $t+7$ in section (4.1). The nominal interest rate follows the same hump shape as before. Intuitively we could think of the delayed response as a consequence of not having the diverging market beliefs creating a monetary policy shock. Thus we get a dampened response of the output gap and inflation to the demand shock.

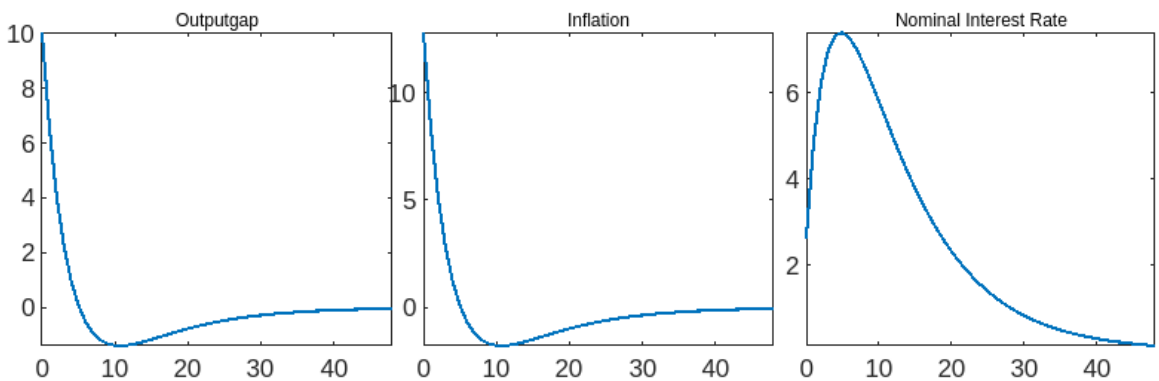

Figure 4: Impulse Response Functions For The Common Beliefs Model with High Persistence Calibration

Comparing these plots with the empirical results in Christiano et al., 2005, we can see that the drop just below 0 before converging to the steady state is more coherent with the data. However we still do not have the initial lags and positive hump-shaped responses for both π_t and $\widetilde{y}_t$ we see in the data.

In the low calibration version in Figure 5 we see almost no differences in the IRFs for $\widetilde{y}_t$, π_t and i_t. In equation (8) we stated that $\psi + \omega_s + \omega_g = 1$. A lower value of the persistence psi implies a higher ω_s and, or ω_g. Even if they disagree, agents place a higher weight on observed data in the current period $s_t + \mu_t^j$ as well as in the actual outcome for demand g_{t-1}. They will change their beliefs according to the facts. As

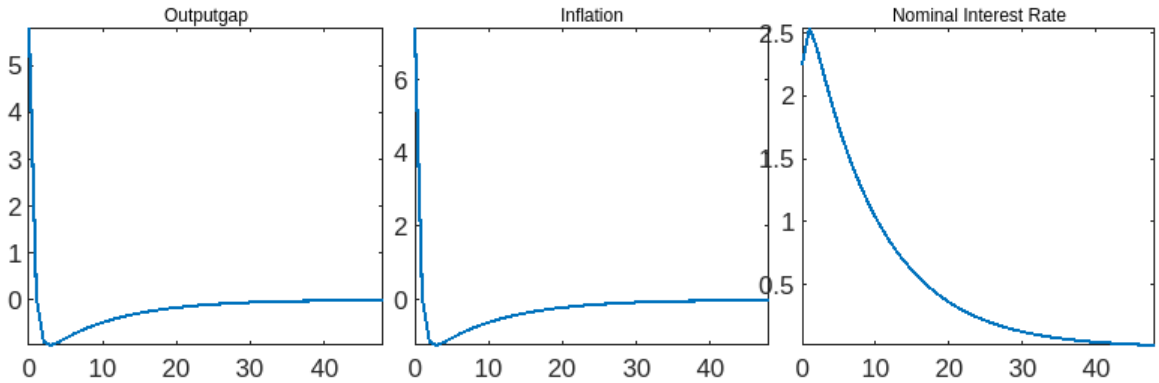

Figure 5: Impulse Response Functions For The Common Beliefs Model with Low Persistence Calibration

such, we expect agents to have the same beliefs about structural aggregate demand G_t much faster. It makes sense that the low calibration IRFs closely resemble the common beliefs with low calibration.

We can conclude that the backwards looking expectation formation assumption for G_t is the main driving force of the model. This assumption is sufficient to generate impulse response functions that that exhibit some of the features of the data (Christiano et al., 2005). Disagreements do offer some persistence and add variance to the reaction of macroeconomic variables. If we are interested in exploring further consequences of disagreements, it is better to have a high value for the persistence parameter ψ. Otherwise our results closely resemble the no disagreements scenario. However we are still unable to generate the lagged responses for the nominal interest rate, output and inflation following a monetary policy shock.

4.4 Potential Consequences for Central Bank Communication

We haven't specifically addressed central banking communication in our model. However we can still try to withdraw some normative conclusions for the role of commu-

nication in this framework.

In the original paper of Caballero and Simsek, 2022 both agents agree to disagree. The central bank does not try to persuade the market to agree to its view. However communication could help the central bank to reduce the persistence ψ of private agents expectation formation. This would not prevent disagreements from arising in the first place. That is an exogenous process. However it could help disagreements to subside quicker and converge to common beliefs. This would reduce the persistence and fluctuations we see in inflation and the output gap. As demonstrated by McMahon and Rholes, 2023, survey data shows that agents display decency bias in their expectations formation. As such, our analysis could be helpful to design a communication strategy that takes into account this updating behaviour.

Furthermore, the persistence that disagreements introduce in the model is a desired property. Monetary policy is less powerful than in the New Keynesian model. Classical iterations of the New Keynesian model often overestimate the power of monetary policy. This feature could be useful to solve the forward guidance puzzle (see Del Negro et al., 2012). At the zero lower bound, central bank communication aims to increase long term interest rates through forward guidance by influencing the expectations of financial market participants. Our analysis could potentially help to understanding the why the financial markets channel makes forward guidance at the zero lower bound less effective than what theory predicts.

5 Limitations and Further Steps

The first limitation of our analysis is that we adopted a simplified version of the Caballero and Simsek, 2022 model. We used the first order conditions provided in the paper and plugged them into dynare to derive impulse response functions. In

doing so, we simplified the process that generates the interpretation of the demand for each agent $s_t + \mu_t^j$ in equation (8). This might have contributed to some of the counter-intuitive impulse response functions we derived.

We also stated without proof the original model under different assumptions. We combined assumptions in section (3.2)-(3.3) with the original first order conditions of Caballero and Simsek, 2022, without any formal proof that the different assumptions yield the same first order conditions. It might be the case that some of the assumptions lead to entirely different first order conditions and model behavior. We leave such proofs to future work. Similarly we adopted the original calibrations of the paper for each iteration of the model. A more careful and tailored calibration of the model parameters for each iteration might help us get rid of the contradictory results.

Another limitation is that we have assumed that the market and the central bank have homogeneous parameters in their weighted average composition of beliefs in equations (8) and (17). However in the real world, both asset markets participants and policymakers form expectations under different incentives. We have no strong reason to believe they would assign the same weight to each parameter. We could conduct data analysis to better calibrate the persistence of old beliefs for each agent. We could also think of a model that presents two different belief formation systems for each agent in order to to represent this potential heterogeneity.

Furthermore, Caballero and Simsek, 2022 were interested in explaining the differences between the Federal Reserve's expected future interest rate path and that of the market's. According to empirical studies, these do diverge sometimes (Couture, 2021). The authors focus on how interest rate expectations adjust and also on the micro foundations of monetary policy shocks. Most of the results with market expectations are derived under a zero inflation scenario ($\kappa = 0$ in equation (1)). The

authors also fail to provide any discussion on the consequences of disagreements for the output gap and inflation rate beyond deriving their closed form expressions in the last sections. As such, the model may not be well suited to think about how backward-looking heterogeneous expectations impact aggregate macroeconomic variables.

Regarding monetary policy, the conclusions of our analysis contradict Caballero and Simsek, 2022's conclusions. The authors show that disagreements lead to monetary policy gradualism. In other words, in response to a positive demand shock, if the market is less optimistic about demand than it is, it will increase interest rates gradually to achieve its target inflation. In most variations of the model, the nominal interest rate immediately reacts in the period that shock occurs. It reaches its maximum is at most 5 periods after the shock. It then gradually converges back to steady state. These result seems at odds not only with Caballero and Simsek, 2022 but also with what we observe in monetary policy in the real world. For instance the Federal Reserve increases interest rates in increments of $0.25 - 0.75$ basis points. An sudden overnight increase in period t that surprises the market is not coherent with monetary policy as we know it currently.

Moreover the conclusions for monetary policy communication are scarce. Caballero and Simsek, 2022's model does not use central bank communication to persuade the market. Communication ensures the market does not misinterpret the central bank's beliefs about the state of demand. In our analysis communication does have role in the instances where disagreements do arise. However the conclusions remain purely normative. From a modeling perspective, no improvements are made with regards to central bank communication in the context of the New Keynesian model. Perhaps another specification of the model could provide a better role for central bank communication.

An interesting next step would be to evaluate the model with different types private sector expectations. In our model we assumed the market fulfill the role of the households in the classic New Keynesian model. Thus their expectations impact our general equilibrium result through the IS curve (equation (3)). We also assumed their expectations drive the Phillips curve (equation (1)). To obtain this result the model makes an implicit assumption that the market is also a price setter. Thus it also plays the role of the firm. There has been some recent empirical studies on non-market participants beliefs and how they affect monetary policy (for an example see Ehrmann and Wabitsch, 2022). We could design a model where households and firms have different types of opinions on aggregate demand based on their degree of knowledge of financial markets. Market participants own equity and bonds so their expectations could be represented through the firm channel. Households could inherit the non-expert beliefs. We leave this idea for another paper.

6 Conclusion

We have shown that the new assumptions about expectation formation of financial markets and the central bank provide new insights to the New Keynesian model. We found that allowing for disagreements between the two generates persistence that is not originally present in the New Keynesian model. However besides a slower convergence rate, the IRFs are not any different from the classical New Keynesian model. The backwards-looking Bayesian expectations formation mechanism is responsible for the hump-shaped IRFs. These mark a departure from the canonical model and provide a closer fit to the data. Nevertheless, we are still far from a close match with empirical findings. Inflation output and the nominal interest rate do not display the lagged responses to a shock we find empirically. Furthermore, the low persistence

calibration of the baseline model exhibits very little differences from the our common beliefs version. To explore disagreements we require high persistence of past beliefs. Some of the IRFs for the expectations of both agents seem counter-intuitive and need further analysis.

These findings offer a potential role for monetary policy. Central bank communication can help to reduce the persistence of previous market beliefs. This would reduce the variability in the responses of inflation and the output gap to monetary policy. Similarly the persistence in the IRFs is a desirable property. In most New Keynesian models, monetary policy communication is too powerful, particularly in the case of forward guidance at the zero lower bound. In our models, after a shock, inflation and output slowly converge to the steady state. We can thus conclude that even if the results are not perfect, the new assumptions do provide more realistic results within the framework of the New Keynesian model. Central bank communication can have a role in the model. We leave further modelling developments and calibrations to future papers.

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